price per gm in Mumbai market. The actual observations given in the table indicates that the gold price ranges from ₹ 2893 to ₹ 2927 in Chennai market and the gold price range from ₹ 2895 (per gram) to ₹ 2925 (per gram) in Mumbai market. Further it is to be noted that there is some oscillations in the gold price rate dated between 20th Feb to 24th Feb and 25th Feb price is remain same as the previous day that is Feb 24th. It may be due to holiday of Gold markets. It is clear from the above table the price of gold rate go on rapidly decreasing from 27th Feb to 1st March. The same fluctuations is observed in Mumbai market from Feb 20th to 24th and remain same on 24th and 25th and rapidly decreasing from Feb 27th to 1st March . So, the market trend in respect of gold rate is same in two markets. We found that the average gold price in Chennai market during these 10 days is ₹ 2916.4 (per gm) and the Mumbai market is ₹ 2913.8 (per gm). The variation among the prices of gold in 10 days. To identify the variation of each individual observation from the mean of all observations. We make use of the best measure known as standard deviation. In this study it is found that the price of gold has a average deviation of ₹ 10 (approximately) in Chennai market and ₹ 9 (approximately) in Mumbai market. To verify the consistency of the prices between the two cities, we attempt coefficient of variation which expresses the percentage of variation. Coefficient of variation of gold price in Chennai market is

$$CV_X = \frac{\sigma_X}{\overline{X}} \times 100 = \frac{10}{2916.4} \times 100 = 0.343\%$$

Similarly, Coefficient of variation of gold price in Mumbai market is

$$CV_Y = \frac{\sigma_Y}{Y} \times 100 = \frac{9}{2913.8} \times 100 = 0.31\%$$

Comparison of these coefficients of variations we inferred that the Mumbai market has consistent or more stable in price of gold.

Further to examine the linear relationship between the two variables X and Y. We carry out correlation analysis results $r(X,Y) = 0.8682$. It indicates that there is a positive correlation in price of gold between Mumbai market and Chennai market. Do you think finding a regression line makes sense here?

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